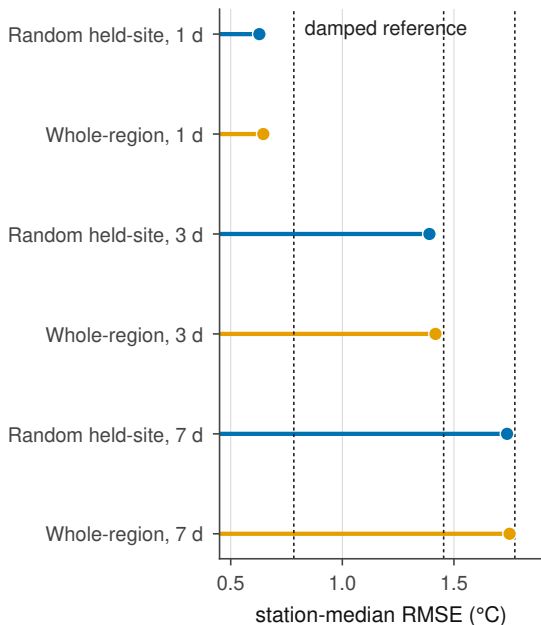
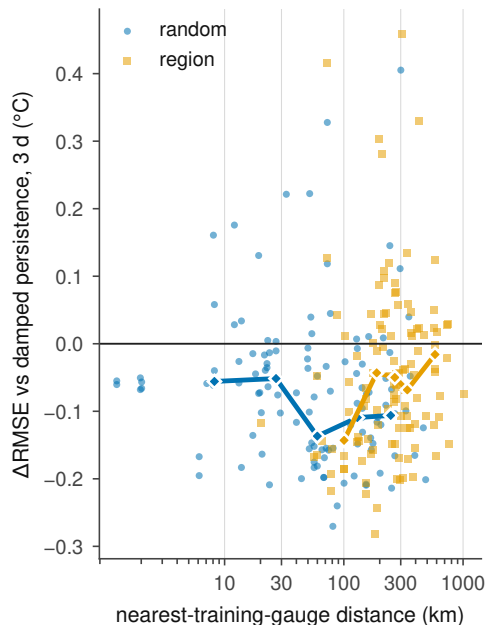


Matched spatial transfer, independent 2021–2023 window

(a) Spatial arms, 2021–2023



(b) Transfer penalty by distance



Station-agnostic LightGBM with frozen per-lead hyperparameters (one seed); four deterministic whole-HUC2-region folds versus four balanced random folds; identical preprocessing in both arms; unweighted station medians over the 118 scored sites. Median nearest-training-gauge distance: 268 km (region) versus 60 km (random).